

Ayush Chauhan

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Education

Chitkara University

Bachelor of Engineering in Computer Science and Engineering; CGPA: 9.31

Rajpura, Punjab, India

Expected May 2028

Experience

Google Summer of Code – OpenPrinting | *Software Engineering Intern*

May 2026 – Present

- Building a CPDB-based print dialog frontend for the COSMIC desktop in Rust and libcosmic, providing native printing workflows for Linux desktop and Flatpak applications
- Integrated zbus into cpdb-rs, replacing glib-based IPC with asynchronous D-Bus support and enabling compatibility with pure async Rust environments
- Authored **5+** code examples illustrating async D-Bus integration and CPDB workflows for Rust applications

Turbin3 Builder Cohort | *Solana Ecosystem Developer*

Jan. 2026 – Mar. 2026

- Built and deployed **5+** smart contract programs on Solana using Anchor – vaults, escrow systems, AMM liquidity pools, NFT staking, and a provably fair dice game with on-chain ed25519 signature verification
- Designed Samizdat Protocol, a decentralized digital signage DePIN platform, applying distributed systems software design to a peer-to-peer infrastructure capstone

LF Decentralized Trust | *Open Source Mentee*

Jun. 2025 – Nov. 2025

- Ported an MCP NTP Timeserver to a TMCP service, enabling trusted time verification for AI agent ecosystems
- Designed 2 JSON-RPC protocols for credential issuance and presentation with replay protection, selective disclosure, and P-256 cryptographic operations
- Developed a modular PoW-based DID credential framework and validated interoperability with open-source LLMs including Qwen and Llama

Projects

Raven | *Rust, Tokio, Tonic, Axum, ClickHouse, VictoriaMetrics* | github.com/rvnhq/raven

- Engineered a Rust observability platform comprising a Linux agent, gRPC ingestion service, and Axum control plane, storing logs in ClickHouse and metrics in VictoriaMetrics with a React dashboard for visualization
- Implemented WAL-backed buffering, TLS-secured transport, inotify log tailing, and procfs-based system sampling through dedicated telemetry collection modules
- Sustained **41K+** HTTP requests/sec and **900+** gRPC RPCs/sec on a single container instance during concurrent load testing

Gitx | *Go, Bubbletea, Charm Stack, GitHub Actions* | github.com/gitxtui/gitx

- Created a terminal-based Git client in Go supporting staging, commit history inspection, branching, stash management, and repository navigation
- Implemented customizable keybindings, themes, and workflow enhancements across **10 public releases** based on user feedback and bug reports
- Distributed cross-platform binaries for Linux, macOS, and Windows, reaching **200+ downloads** while coordinating contributions from **8 developers**

lq | *Go, CLI, GitHub Actions* | github.com/bakayu/lq

- Built a CLI utility that generates LICENSE and .gitignore templates for new repositories in a single command
- Added support for all GitHub-recognized open-source licenses and templates covering popular programming languages and frameworks

Community and Open Source

Technical Lead & Core Team | *Open Source Chandigarh*

Sep. 2024 – Present

- Leading technical direction and architecture for community projects; Delivered Git, GitHub, and Linux workshops introducing **150+** participants to command-line tooling and open-source development practices

Open Source Contributor | *Graphite, OpenWallet Foundation, OpenPrinting, Layer5io*

Aug. 2024 – Present

- Contributed to open-source organizations with work spanning procedural 2D vector graphics, developer tooling, identity infrastructure, and Linux printing subsystems; maintained public CLI tools and backend libraries

Technical Skills

Languages: Rust, Golang, Java, Python, C, TypeScript, SQL (PostgreSQL)

Systems and Backend: Tokio, Axum, Actix-web, Tonic (gRPC), Gin, async runtimes, D-Bus (zbus), IPC, inotify

DevOps and Tools: Git, Docker, Docker Compose, GitHub Actions, Kubernetes, Azure, Bash, Linux